

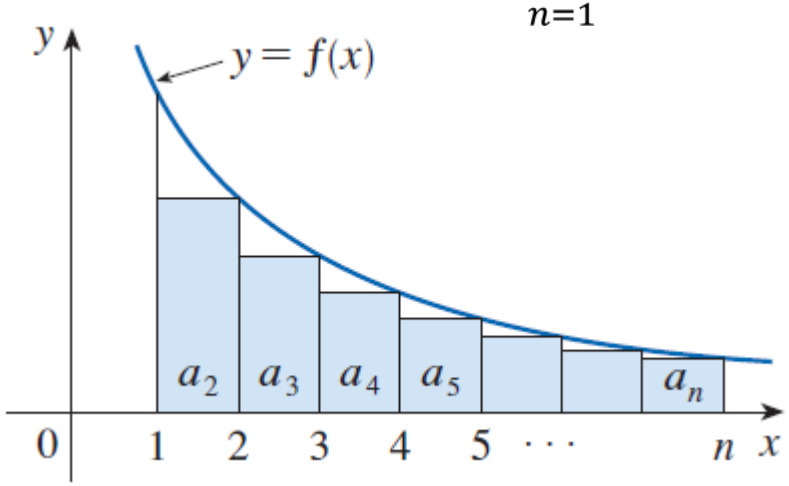
Chapter 9:

Infinite Series

Section 9.3: The Integral Test and p-Series

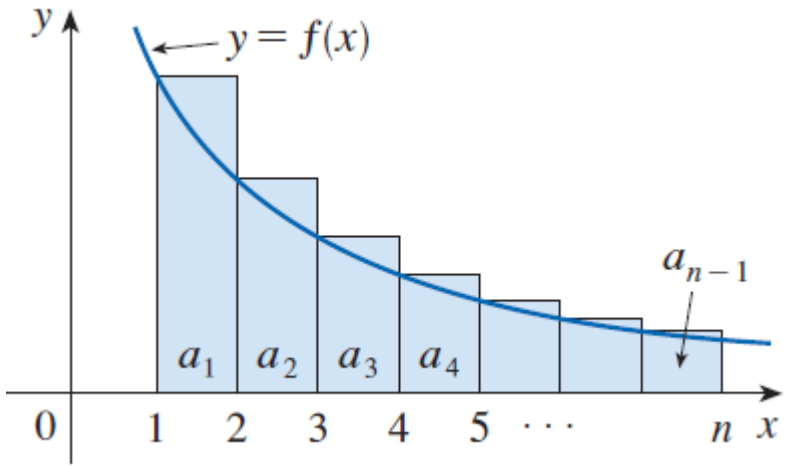
Consider the series

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + a_4 + a_5 + \ \ldots$$,where the sequence $\{a_n\}$ is positive and decreasing.



$$a_2 + a_3 + \cdots + a_n \leqslant \int_1^n f(x)\,dx$$

$$s_n = a_1 + \sum_{i=2}^{n} a_i \leqslant a_1 + \int_1^{\infty} f(x)\,dx$$



$$\int_1^n f(x)\,dx \leqslant a_1 + a_2 + \cdots + a_{n-1}$$

$$\int_1^n f(x)\,dx \leqslant \sum_{i=1}^{n-1} a_i = s_{n-1}$$

The Integral Test Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$. Then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the improper integral $\int_1^{\infty} f(x)\,dx$ is convergent. In other words:

(i) If $\int_1^{\infty} f(x)\,dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.

(ii) If $\int_1^{\infty} f(x)\,dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

To use The Integral Test, we must check that the function f is:
1) Continuous
2) Positive
3) Decreasing

on the interval $[1, \infty)$.

Remember that the convergence or divergence of $\sum a_n$ is not affected by deleting the first N terms. Similarly, when the conditions for the Integral Test are satisfied for all $x \geq N > 1$, you can simply use the integral $\int_N^{\infty} f(x)\,dx$ to test for convergence or divergence. (This is illustrated in Example 4.)

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=2}^{\infty} \frac{\ln n}{n}$$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1}$$

SOLUTION

p-Series and Harmonic Series

A series of the form

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots$$

p-series

is a ***p*-series,** where p is a positive constant. For $p = 1$, the series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots$$

Harmonic series

is the **harmonic series.**

THEOREM 9.11 Convergence of *p*-Series

The p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \cdots$$

converges for $p > 1$ and diverges for $0 < p \leq 1$.

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{1}{n^{1.3}}$$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=9}^{\infty} \frac{1}{n^2}$$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

SOLUTION

NOTE We should *not* infer from the Integral Test that the sum of the series is equal to the value of the integral. In fact,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \qquad \text{whereas} \qquad \int_1^{\infty} \frac{1}{x^2}\,dx = 1$$

Therefore, in general,

$$\sum_{n=1}^{\infty} a_n \neq \int_1^{\infty} f(x)\,dx$$

EXAMPLE

Conditions of the Integral Test **In Exercises 25–28, explain why the Integral Test does not apply to the series.**

27. $\displaystyle\sum_{n=1}^{\infty} \frac{2 + \sin n}{n}$

SOLUTION

EXAMPLE

Finding Values **In Exercises 45–50, find the positive values of p for which the series converges.**

45. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}$

SOLUTION

Solve exercises of section 9.3.

Start with the suggested problems mentioned in the syllabus.